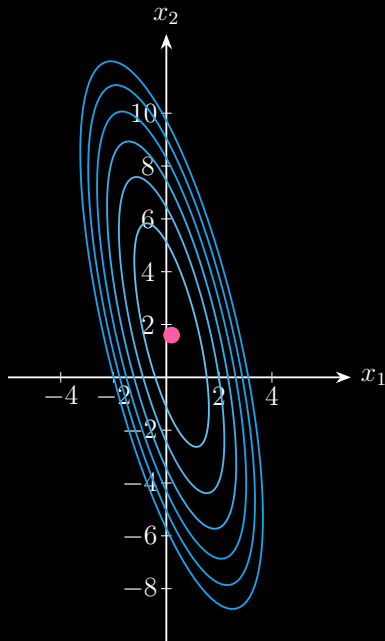


Quadratic optimization:

$$\min_x \quad x^\top A x - b^\top x$$

with

$$A = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$



Quadratic optimization:

$$\min_x \quad x^\top A x - b^\top x$$

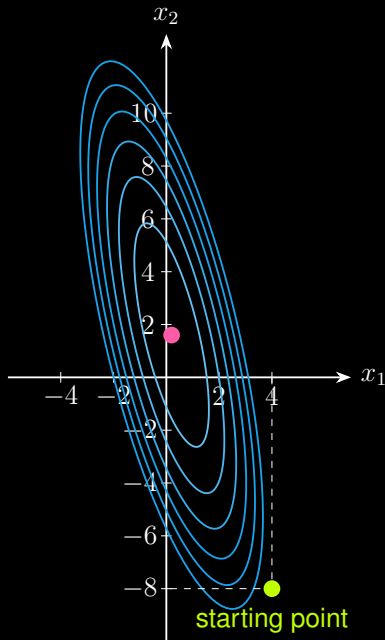
Steepest gradient descent:

$$\mathbf{g}_k = A\mathbf{x}_k - \mathbf{b} \quad (\text{gradient})$$

$$\alpha_k = \frac{\mathbf{g}_k^\top \mathbf{g}_k}{\mathbf{g}_k^\top A \mathbf{g}_k} \quad (\text{step size})$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{g}_k \quad (\text{decision vector})$$

with an initial point $\mathbf{x}_0$.



Quadratic optimization:

$$\min_x \quad x^\top A x - b^\top x$$

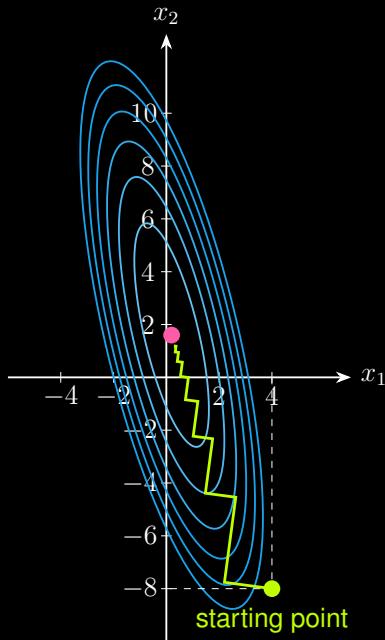
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Thanks for your attention!

About me:

🏠 Homepage: <https://xinychen.github.io>

✉ How to reach me: chenxy346@gmail.com